Signatures

BPDRPT INT-017 Purification Process 2 Description

Contributors to the preparation of this document

Preparer(s) [redacted]

Approver(s) [redacted]

Check th [redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

Function/ Department	Responsibilities
Prepared by: Purification Process Sciences Tech Transfer	Responsible for identifying and/or defining all parameters and pertinent information for transfer to manufacturing as well as the document structure and formatting • Identify all required parameters for transfer to manufacturing based on technology transfer experience to ensure appropriate level of process definition • Maintain consistency across projects and process descriptions • Compile summary tables: BOM, Hold Times, Filtration, Buffers • Verify that targets and ranges for all parameters are defined and are reasonable to maintain process performance within the expected capabilities of facilities
Prepared by: Purification Process Sciences	Responsible for identifying and defining all project specific parameters and values for transfer to manufacturing • Identify required parameters and define values for targets and ranges for all project specific parameters including buffer recipes based on development experience to ensure appropriate product quality and process performance • Compile process updates and justifications for variations relative to platform process (ED000) or prior process description edition in the Process Changes Section
Reviewed by: Manufacturing Sciences Scale-up Center/Scale-up Lab	Responsible for reviewing parameters and verifying that they are representative of typical performance during development and pilot scale • Verify that the same parameters and values, including buffer recipes, were used during scale-up • Verify that parameter targets meet process requirements (i.e. Equilibration Volume target is sufficient to meet pH range)
Approved by: Formulation Sciences	Responsible for reviewing all formulation specific parameters and verifying that they are accurate and in alignment with current FS strategy **Sections:** 1.1, 12-13 • Define final formulation, container, and room temperature hold time • Verify that buffer recipes used during formulation are accurate • Verify that densities are sufficiently accurate
Approved by: Purification Process Sciences	Responsible for reviewing project specific parameters and confirming that they are in alignment with current PPS strategy (DEV0000_CB_0592) **Sections:** 3-15 • Verify that project specific parameter targets and ranges are in alignment with general PPS knowledge and expectations • Verify that variations from platform process and prior process are captured in Process Changes and justified
Approved by: Purification Process Sciences Technology Transfer	Responsible for reviewing that all pertinent information/ parameters for transfer to manufacturing are included and in alignment with current BPE strategy **Sections:** All • Verify that parameter targets and ranges are in alignment with general PPS knowledge and expectations and facility capabilities • Maintain consistency across projects and process descriptions
Approved by: Manufacturing Sciences	Responsible for reviewing that all pertinent information/ parameters for transfer to manufacturing are included and in alignment with current MS strategy as well as providing any MS specific information contained in the document (i.e. part numbers) **Sections:** All

Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

TABLE OF CONTENTS

1 INTRODUCTION 4
1.1 Purpose 4
1.2 Assumptions 5
1.3 Abbreviations/Acronyms/Definition 6
1.4 Definitions of Terminologies for Parameter Ranges 8
1.4.1 Target 8
1.4.2 Normal Operating Range (NOR) 8
2 BILL OF MATERIALS 9
3 IN-PROCESS HOLD TIMES 11
4 LOAD AND PRODUCT FILTRATION 12
5 AFFINITY CHROMATOGRAPHY 14
6 VIRUS INACTIVATION (LOW PH) 16
7 AEX 17
8 AEX PRODUCT ADJUSTMENT 19
9 CEX 20
10 VIRUS FILTRATION 22
11 CONCENTRATION AND DIAFILTRATION 23
12 FORMULATION 26
13 BULK FILTRATION 27
14 COLUMN PACKING AND TESTING 28
15 BUFFERS AND SOLUTIONS 29
16 REFERENCE CHROMATOGRAMS 34
17 PROCESS CHANGES 39
18 EDITION HISTORY 45

Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

1 INTRODUCTION

1.1 Purpose

This report describes the purification process and process parameters for production of INT-017 drug substance. The overall process (upstream/downstream) is designated “Process 2”.

Table 1 General Properties: INT-017 Antibody Intermediate

Property	Description
Target	EGFR-Cmet
Protein classification	Duet mAb Bispecific
Isoelectric point (pI) range (experimental)	7 bands between 8.5 – 9.2 (DLIMS:1271613)
Extinction coefficient (theoretical)	1.51 (theoretical)
Expected titer range (g/L)	4.0 (range: 3.5-4.5)
Overall expected yield from recovery through antibody intermediate (%)	55
Antibody intermediate concentration (g/L)	50
Drug substance buffer	20 mM histidine/histidine-HCl, 240 mM sucrose, pH 6.0
Drug substance presentation	Stored frozen at ≤ - 35°C in CryoVaults

Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

1.2 Assumptions

- This report does not consider scale or facility constraints.
- Unless otherwise noted, procedures pre/post common unit operations should be followed for filters and unit operations per DEV000_GB_0469/REP-0414101.
- Unless otherwise noted, all conductivity values are compensated to a reference temperature of 25°C using a temperature coefficient (θ) of 2.0% per °C.
- The target volumes are expected to be sufficient. Overage to account for system hold up volume is site dependent and should be a separate phase and not added into the volume targets for the process.
- Unless otherwise noted, flow rate ranges are for steady state operation and do not consider pump ramping strategy.
- Unless otherwise noted, all unit operations are conducted at controlled room temperature (15-27°C).
- Unless otherwise noted, buffer preparation may be conducted at higher temperatures as appropriate (15-50°C).
- For facilities for which 1.0 N NaOH is not feasible, 0.5 N NaOH may be used as a substitute with “Conductivity (mS/cm @ 25°C), final” NOR: 84 – 113
- Unless otherwise noted, all chromatography step phases are operated in down flow mode.
- Facility can adjust target and tighten ranges for control as appropriate
- Unless otherwise noted, all process parameters are inputs.

Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

1.3 Abbreviations/Acronyms/Definition

Table 2 Abbreviations and Acronyms

Acronym	Definition
AEX	Anion exchange chromatography
AI	Antibody intermediate
AL	Alert limit
Bldg	Building
BP	British Pharmacopeia
BR	Batch record
CEX	Cation exchange chromatography
CM	Conditioned media
Conc.	Concentration
CV	Column volume
DV	Diavolumes
DBC	Dynamic binding capacity
DCF	Density Correction Factor
DS	Drug Substance
DSC	Development Scale-Up Center (500-L scale)
EAB	Excipient addition buffer
ELISA	Enzyme-linked immunosorbent assay
EU	Endotoxin Unit
EP	European Pharmacopeia
FIO	For information only
FF	Fast Flow
FMC	Frederick Manufacturing Center
FSS	Formulation Spiking Solution
GPF	Gaithersburg Pilot Facility
GMP	Good Manufacturing Practices
HDPE	High Density Polyethylene
HETP	Height equivalent to a theoretical plate
HIC	Hydrophobic interaction chromatography
HPLC	High-performance liquid chromatography

Acronym	Definition
LG	Linear gradient
LR	Lot release
LMH	Liters/meter^2-hour
MM	Mixed mode chromatography
MV	Membrane volume
MWCO	Molecular Weight Cut Off
NA	Not applicable
NC	Non-conformance
NF	National Formulary
NOR	Normal operating range
OD	Optical density
PCR	Polymerase chain reaction
PES	Polyethersulfone
PF	Product Filter
PFI	Pre-formulated intermediate
P/N	Part number
PP	Polypropelyne
psid; psig	pounds per square inch differential; pounds per square inch gauge
Protein A	Protein A affinity chromatography
qs	Bring to final weight (quantum satis)
RPHPLC	Reverse phase high performance chromatography
S.F.	Safety factor
SOP	Standard Operating Procedure
SS	Stainless steel
SUL	Scale-Up Lab
TBD	To be determined
TMP	Trans-membrane pressure
UB	Unfiltered Bulk
UDS	Unformulated Drug Substance

Check th version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

Table 2 Abbreviations and Acronyms

Acronym	Definition	Acronym	Definition
HPSEC	High-performance size-exclusion chromatography	USP	United States Pharmacopeia
HCP	Host cell protein	VC	Viral clearance
IgG1/2/4	Immunoglobulin G1/2/4	VF	Virus filtration
IEC	Ion exchange chromatography	VPro	Viresolve Pro
JP	Japanese Pharmacopeia	WFI	Water for injection
kDa	kilodalton		
LF	Load Filter		

Check th version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

1.4 Definitions of Terminologies for Parameter Ranges

1.4.1 Target

The target value is a preset value that a process parameter is aimed to by process controls. The controls include any combination of manual or automated, discrete or continuous, feed-forward or feed-back responses to operate the process near the target value. Where Targets are provided, the process should be controlled to meet the target value as closely as possible, not just controlling to the edge of the Normal Operating Range.

1.4.2 Normal Operating Range (NOR)

Operating parameter ranges, also called Action Limits, which are established based on current understanding and process knowledge gained through development and process characterization. NORs are specified in manufacturing batch records and/or other GMP documentation (such as automation recipes) to control the process. A value exceeding a NOR triggers a non-conformance investigation and is documented and evaluated according to the site's Quality System.

Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

2 BILL OF MATERIALS

Table 3 Bill of Materials

Type	Component Name	Minimum Grade	MEDI P/N	Justification/Notes
Chromatography media	CaptureSelect Lambda XP (Affinity)	NA	110033212 110033958	5 L 10L
	Capto Q (AEX)	NA	6001609 6001621	5 L 10 L
	Nuvia HR-S (CEX)	NA	6002211	10 L
Buffer and solution components	Acetic acid, glacial	USP, EP	4101341 6000325	NA
	Benzyl alcohol	NF, EP, JP	4102670	NA
	L-Histidine	EP, JP	4101095	NA
	L-Histidine HCl monohydrate	JP, EP	4101011	NA
	Sodium acetate, trihydrate, crystal	USP, EP, JP	4101342 6000326	NA
	Sodium chloride	USP, EP	6000324	NA
	10N Sodium hydroxide	NA	4102099 4102229	Biotech Reagent
	Sodium hydroxide, pellets	EP, NF	4101097	NA
	Sodium Phosphate, Dibasic, Heptahydrate	USP, EP	4101092 6003286	NA
	Sodium Phosphate, Monobasic, Monohydrate	USP, EP	4101091	NA
	Sucrose (low endotoxin)	USP, BP, JP	4102624	NA
	Tris base (tromethamine/trometamol)	USP, EP	4101399	NA
	Tris hydrochloride	NA	4100924	Biotech Reagent
Purification filters	Millistak+ HC depth Pod filters - X0HC	NA	4101860	1.1 m^2
	Millistak+ HC depth Pod filters - C0HC	NA	4101552	1.1 m^2
	3M Zeta Plus filters	NA	6005228 6005229	1.6 m^2 (Standard) 1.6 m^2 (Alkaline resistant)
	Millipore Express SHC, 0.5/0.2 µm	NA	NA	Generic 0.2 µm
	Pall Fluorodyne EX EDF, 0.2 µm	NA	NA	Generic 0.2 µm
	Viresolve Pro Magnus Device	NA	4102501 4102503	0.51 m^2 1.53 m^2
	Viresolve Pro Magnus Shield	NA	4102530 4102531	0.51 m^2 1.53 m^2

Check th version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

Table 3 Bill of Materials

Type	Component Name	Minimum Grade	MEDI P/N	Justification/Notes
	Pellicon 3 Ultracel C Screen ultrafiltration cassettes, 30kDa MWCO	NA	6001576 4102051 4102050	0.11 m^2 0.57 m^2 1.14 m^2
	Durapore Opticap, 0.2 µm	NA	NA	Sterile Grade 0.2 µm
Miscellaneous components	Meissner CryoVaults	NA	6003782 6003895	20L

Check th version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

3 IN-PROCESS HOLD TIMES

Process intermediate stability studies were performed to support pooling of chromatography cycles and holding of process intermediates.

Hold times begin when product is collected and end when the next step is loaded.

Unless otherwise noted, room temperature hold times are inclusive to refrigerated temperature hold times but not vice versa. For example, a product that was held for 48 hours at room temperature prior to being stored 24 hours refrigerated would have a total hold time of 72 hours at refrigerated temperature but only 48 hours at room temperature.

Table 4 Process Intermediate Hold Times

Process Intermediate	Room Temperature Hold Time - Max (days, 15-25 °C)	Refrigerated Temperature Hold Time - Max (days, 2-8 °C)	Justification/Notes
Conditioned medium	7	14	DEV-20240314-001
Lambda XP Product	6	14	DEV-20240410-003
Virus Inactivation Product, pre-filter	6	14	
Virus Inactivation Product, post-filter (if applicable)	14	14	
AEX Product	8	14	
CEX Load Adjustment Product	14	14	
CEX Product	14	14	
Virus Filtration Product	14	14	Leverages CEX Product Data
Pre-Formulated Intermediate	14	14	DEV-20240410-003
Unformulated Bulk	2	7	DEV-20211117-005 DEV-20220502-001

Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

4 LOAD AND PRODUCT FILTRATION

Table 5 Filtration Parameters

Unit Operation	Position	Membrane Type	In-line	Volumetric Load NOR-Max (L/m^2)	Justification/Notes
Affinity Chromatography (LF 1 → LF 2 → Unit Op → PF)	Load Filter 1	Millistak+ HC POD filter - X0HC	Load – Equil 2	587	Offline Pmax
		3M Zeta Plus filter	Load – Equil 2	728	Offline Pmax
	Load Filter 2	Express SHC, 0.5/0.2 µm (With X0HC)	All Steps	3857	Offline Max challenge (with SHC in line with XOHC)
		Express SHC, 0.5/0.2 µm (With 3M)	All Steps	5192	Offline Max challenge (with SHC in line with 3M)
	Product Filter	Express SHC, 0.5/0.2 µm	Collection	3000	NA
Virus Inactivation (Unit Op → PF)	Product Filter 1	Millistak+ HC POD filters – C0HC	Off-line	776	Offline Max challenge is 776 L/m2 for lot SP24-038 with minimal pressure increase observed
	Product Filter 2 (If applicable)	Express SHC, 0.5/0.2 µm	Off-line	5099	Offline Vmax (with SHC in line with COHC)
AEX Chromatography (LF → Unit Op → PF)	Load Filter	Express SHC, 0.5/0.2 µm	All Steps	5099	Offline max challenge is 5099 L/m2 for lot SP24-038 with minimal pressure increase observed
	Product Filter	Express SHC, 0.5/0.2 µm	Collection	3352	Offline max challenge is 3352 L/m2 for lot SP24-038 with minimal pressure increase observed

Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

Table 5 Filtration Parameters

Unit Operation	Position	Membrane Type	In-line	Volumetric Load NOR-Max (L/m^2)	Justification/Notes
CEX Chromatography (LF → Unit Op → PF)	Load Filter	Express SHC, 0.5/0.2 µm	All Steps	3000	Max challenge at SUL is 1378 L/m2 for lot SP24-038 with minimal pressure increase observed
	Product Filter	Express SHC, 0.5/0.2 µm	Collection	3519	Offline max challenge is 3519 L/m2 for lot SP24-038 with minimal pressure increase observed
Virus Filtration (Unit Op → PF)	Load Filter	Express SHC, 0.5/0.2 µm	All Steps	3519	Offline max challenge is 3519 L/m2 for lot SP24-038 with minimal pressure increase observed
	Product Filter	Express SHC, 0.5/0.2 µm	All Steps	3000	Max challenge at SUL is 1030 L/m2 for lot SP24-038 with minimal pressure increase observed
Concentration and Diafiltration (Unit Op → PF)	Product Filter	Express SHC, 0.5/0.2 µm	Off-line	182	Recommend smallest device within range. This is the Vmax value obtained at development.
Antibody Intermediate Formulation and Bulk Filtration (Unit Op → Bulk Filter)	Bulk Filtration	Durapore Opticap, 0.2 µm	Off-line	1902	Recommend smallest device within range. Offline max challenge with minimal pressure increase observed

Millipore SHC filters may be interchanged with Pall Fluorodyne EX EDF filters of the same device size. This assumes the additional filter area of the Pall device compensates for any difference in maximum volumetric load.

Check th version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

5 AFFINITY CHROMATOGRAPHY

Step	Process Parameter (units)	Target	NOR	Justification/Notes
General	Media	CaptureSelect Lambda XP		Prep per appropriate SOP.
Equilibration 1 *50 mM tris,* *pH 7.4*	Linear velocity (cm/hr)	200	≤ 220	NA
	Volume (CV)	3.0	≥ 2.5	NA
	pH, effluent, (endpoint check) [a]	NA	7.0 – 7.8	NA
	Conductivity, effluent, (endpoint check) (mS/cm @ 25°C) [a]	NA	NA	Run and record.
Load *Conditioned Medium*	Linear velocity (cm/hr)	200	≤ 220	NA
	Protein load (g/L resin)	NA	15 – 30	NOR-Max is the capacity.
Equilibration 2 *50 mM tris,* *pH 7.4*	Linear velocity (cm/hr)	200	≤ 220	NA
	Volume (CV)	3.0	≥ 2.5	NA
Wash *50 mM tris,* *0.5 M sodium chloride,* *pH 7.4*	Linear velocity (cm/hr)	200	≤ 220	NA
	Volume (CV)	3.0	≥ 2.5	NA
Equilibration 3 *50 mM tris,* *pH 7.4*	Linear velocity (cm/hr)	200	≤ 220	NA
	Volume (CV)	3.0	≥ 2.5	NA
Elution *75 mM sodium acetate,* *pH 3.8*	Linear velocity (cm/hr)	150	≤ 165	NA
	Product collection start (OD_{280})	0.5	≥ 0.3	Volume before peak = ~ 1.0 CV
	Product collection end (OD_{280})	0.5	≥ 0.3	Product volume = ~1.8 – 2.2 CV Step yield = ~75 – 79 %
Strip 1 *100 mM acetic acid*	Linear velocity (cm/hr)	200	≤ 220	NA
	Volume (CV)	3.0	≥ 2.5	NA

Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

Step	Process Parameter (units)	Target	NOR	Justification/Notes
WFI Flush *WFI*	Linear velocity (cm/hr)	200	≤ 220	NA
	Volume (CV)	1.0	≥ 0.5	NA
Strip 2 *25 mM sodium hydroxide*	Linear velocity (cm/hr)	200	≤ 220	NA
	Volume (CV)	3.0	≥ 2.5	NA
	Conductivity, effluent, (endpoint check) (mS/cm @ 25°C) [a]	NA	84 – 113	NA
WFI Flush *WFI*	Linear velocity (cm/hr)	200	≤ 220	NA
	Volume (CV)	1.0	≥ 0.5	NA
Strip 3 *2.0% (v/v) benzyl alcohol, 167 mM acetic acid*	Linear velocity (cm/hr)	200	≤ 220	Omit for last cycle and progress directly to Strip 4.
	Volume (CV)	3.0	≥ 1.9	
	pH, effluent, (endpoint check) [a]	NA	2.5 – 2.9	
End of Cycle				
Strip 4 *2.0% (v/v) benzyl alcohol, 120 mM phosphoric acid, 167 mM acetic acid*	Linear velocity (cm/hr)	200	≤ 220	NA
	Volume (CV)	3.0	≥ 1.9	Sanitisation buffer
	pH, effluent, (endpoint check) [a]	NA	≤ 2.5	NA
Sanitization Hold	Contact time (minutes)	30	29 – 60	NA
Pre-Storage Equilibration *50 mM tris, pH 7.4* OR *WFI*	Linear velocity (cm/hr)	200	≤ 220	NA
	Volume (CV)	1.0	≥ 0.5	NA
Storage *2% (v/v) benzyl alcohol, 100 mM sodium acetate, pH 5.0*	Linear velocity (cm/hr)	200	≤ 220	NA
	Volume (CV)	3.0	≥ 2.5	NA
	pH, effluent, (endpoint check) [a]	NA	4.8 – 5.2	NA

[a] Process output

Check th version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

6 VIRUS INACTIVATION (LOW PH)

Step	Process Parameter (units)	Target	NOR	Justification/Notes
Acid Titration *1 M acetic acid*	pH, (endpoint check) [a]	3.5	3.4 ─ 3.6	~ 0.098 – 0.126 (kg acid / kg Product)
Incubation	Duration (minutes)	65	60 ─ 240	NOR-Min is based on viral clearance recommendation. NOR-Max is based on development stability data.
	Temperature (°C)	NA	18 ─ 25	Confirm product temperature at start and end of incubation
	pH	NA	3.4 ─ 3.6	Confirm pH at start and end of incubation
Base Titration *1 M tris base*	pH, (endpoint check) [a]	7.4	7.2 ─ 7.6	1.751 – 1.532 (kg base / kg acid) Step yield = ~99%

[b] Process output

Check th version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

7 AEX

Step	Process Parameter (units)	Target	NOR	Justification/Notes
General	Media	Capto Q		Prep per appropriate SOP
Pre-Equilibration *50 mM tris,* *0.5 M sodium chloride,* *pH 7.4*	Linear velocity (cm/hr)	300	≤ 330	NA
	Volume (CV)	5.0	≥ 4.5	NA
Equilibration *50 mM tris,* *pH 7.4*	Linear velocity (cm/hr)	300	≤ 330	NA
	Volume (CV)	3.0	≥ 2.5	NA
	pH, effluent, (endpoint check) [a]	NA	7.2 – 7.6	NA
	Conductivity, effluent, (endpoint check) (mS/cm @ 25°C) [a]	NA	NA	Run and record.
Load *Virus Inactivation Product*	Linear velocity (cm/hr)	300	≤ 330	NA
	Protein load (g / L resin)	NA	≤ 100	NOR-Max is the capacity.
	Product collection start (OD_{280})	0.5	≥ 0.3	Volume before peak = ~ 1.0 CV
Chase *50 mM tris,* *pH 7.4*	Linear velocity (cm/hr)	300	≤ 330	NA
	Product collection end (OD_{280})	0.5	≥ 0.3	Product dilution= ~1% Step yield = ~95%
Salt Strip *50 mM tris,*	Linear velocity (cm/hr)	300	≤ 330	NA

Check th version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

Step	Process Parameter (units)	Target	NOR	Justification/Notes
0.5 M sodium chloride, pH 7.4	Volume (CV)	3.0	≥ 2.5	NA
NaOH Strip *1.0 N sodium hydroxide*	Linear velocity (cm/hr)	300	≤ 330	NA
	Volume (CV)	3.0	≥ 2.5	Sanitization buffer
	Conductivity, effluent, (endpoint check) (mS/cm @ 25°C) [a]	NA	164 – 211	NA
End of Cycle				
Sanitization Hold	Contact time (minutes)	35	15 – 120	NA
Pre-Storage Equilibration *50 mM tris, 0.5 M sodium chloride, pH 7.4* **OR** *WFI*	Linear velocity (cm/hr)	300	≤ 330	NA
	Volume (CV)	3.0	≥ 2.5	NA
Storage *2% benzyl alcohol, 100 mM sodium acetate, pH 5.0*	Linear velocity (cm/hr)	300	≤ 330	NA
	Volume (CV)	3	≥ 2.5	NA
	pH, effluent, (endpoint check)[a]	NA	4.8 – 5.2	NA

[c] Process output

Check th version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

8 AEX PRODUCT ADJUSTMENT

Step	Process Parameter (units)	Target	NOR	Justification/Notes
pH Adjustment *1 M acetic acid*	pH	5.0	4.8 – 5.2	~0.06 – 0.072 (kg titrant / kg AEX Product)

Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

9 CEX

Step	Process Parameter (units)	Target	NOR	Justification/Notes
General	Media	Nuvia HR-S		Prep per appropriate SOP
Equilibration 1 *50 mM sodium acetate,* *pH 5.0*	Linear velocity (cm/hr)	150	≤ 165	NA
	Volume (CV)	5.0	≥ 4.5	NA
	pH, effluent, (endpoint check) [a]	NA	4.8-5.2	NA
	Conductivity, effluent, (endpoint check) (mS/cm @ 25°C) [a]	NA	NA	Run and record.
Load *pH Adjusted AEX Product*	Linear velocity (cm/hr)	150	≤ 165	NA
	Protein load (g / L resin)	NA	17 – 27	NOR-Max is the capacity.
Wash *35 mM sodium phosphate,* *pH 6.9*	Linear velocity (cm/hr)	150	≤ 165	NA
	Volume (CV)	3.0	≥ 2.5	NA
Equilibration 2 *50 mM sodium acetate,* *pH 5.0*	Linear velocity (cm/hr)	150	≤ 165	NA
	Volume (CV)	3.0	≥ 2.5	NA
Elution *50 mM sodium acetate,* *200 mM sodium chloride*	Linear velocity (cm/hr)	100	≤ 165	NA
	Product collection start (OD_{280})	0.5	≥ 0.3	Volume before main peak = ~ 1.24 CV

Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

Step	Process Parameter (units)	Target	NOR	Justification/Notes
pH 5.1	Product collection end (OD_{280})	1.5	≥ 1.3	Product volume = ~2.3 – 3.6 CV Step yield = ~75 – 80%
Salt Strip *50 mM Tris, 500 mM sodium chloride, pH 7.4*	Linear velocity (cm/hr)	150	≤ 165	NA
	Volume (CV)	3.0	≥ 2.5	NA
NaOH Strip *1.0 N sodium hydroxide*	Linear velocity (cm/hr)	150	≤ 165	NA
	Volume (CV)	3.0	≥ 2.5	Sanitization buffer
	Conductivity, effluent, (endpoint check) (mS/cm @ 25°C) [a]	NA	164 – 211	NA
End of Cycle				
Sanitization Hold	Contact time (minutes)	35	15 — 120	NA
Storage *0.1 N sodium hydroxide*	Linear velocity (cm/hr)	150	≤ 165	NA
	Volume (CV)	3.0	≥ 2.5	NA
	Conductivity, effluent, (endpoint check) (mS/cm @ 25°C) [a]	NA	19-27	NA

[d] Process output

Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

10 VIRUS FILTRATION

Step	Process Parameter (units)	Target	NOR	Justification/Notes
General	Media	Viresolve Pro +		Prep per appropriate SOP
	Feed pressure (psig)	NA	0 – 60	NA
Equilibration *50 mM sodium acetate,* *200 mM sodium chloride* *pH 5.1*	Differential pressure (psid)	30	0 – 60	NA
	Volumetric load (L/m^2)	75	≥ 65	NA
	pH, effluent, (endpoint check) [a]	NA	NA	Run and record.
Load *CEX Chromatography Product*	Differential pressure (psid)	30	0 – 45	Based on process Acceptable Range maximum of 50 psid per report: *Justification of Upper Pressure Limit for VPro Virus Filter Operation*
	Total volumetric load (L/m^2 filter area)	NA	⩽ 1000	NOR-Max is based on VC study. Measured from the start of product load to the end of post-filtration flush (Equilibration 2)
	Flow decay (%) [a]	NA	≤ 90	100% × (Initial buffer flux - Final virus filtration flux) ÷ (Initial buffer flux)
	Product collection start (OD_{280})	0.5	NA	NA
Chase *50 mM sodium acetate,* *200 mM sodium chloride* *pH 5.1*	Differential pressure (psid)	30	0 – 45	Based on process Acceptable Range maximum of 50 psid per report: *Justification of Upper Pressure Limit for VPro Virus Filter Operation*
	Product collection end (OD_{280})	0.5	NA	Volume chase may be set if OD is not available. Step yield = ~100%

[e] Process output

Check th version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

11 CONCENTRATION AND DIAFILTRATION

Step	Process Parameter (units)	Target	NOR	Justification/Notes
General	Media	Pellicon 3 Ultracel C Screen 30kDa MWCO		Prep per approriate SOP
	Feed pressure (psig)	NA	≤ 60	NA
Equilibration *20 mM Histidine/Histidine-HCl, pH 5.9*	pH, permeate, (endpoint check) [a]	5.9	5.7 – 6.1	NA
Concentration 1 Control Method: Constant TMP and Feed Flow Control	Membrane protein load (g/m^2)	NA	≤ 1000	(Mass of protein)/(area of ultrafiltration membrane) NOR-Max is the tested capacity.
	TMP (psi)	20	≤ 35	NA
	Fed batch ratio	NA	NA	(Load volume)/(DCF tank volume)
	Feed flow ($L/min/m^2$)	5.0	3.0 – 7.0	May lower to maintain constant TMP.
	Protein concentration, endpoint check (g/L)	70.0	NA	May be directly measured or calculated from permeate or retentate volume
Diafiltration *20 mM Histidine/Histidine-HCl, pH 5.9* Control Method: Constant TMP and Feed Flow Control	TMP (psi)	20	≤ 35	NA
	Feed flow ($L/min/m^2$)	5.0	3.0 – 7.0	May lower to maintain constant TMP
	Number of diavolumes (DV)	6.0	≥ 5.5	(kg buffer)/(kg product)
	pH, permeate, (endpoint check) [a]	NA	NA	NA
	pH, retentate, (endpoint check) [a]	NA	5.7 – 6.1	NA
Concentration 2	TMP (psi)	20	≤ 35	NA
	Feed flow ($L/min/m^2$)	NA	≤ 7.0	May lower to maintain constant TMP

Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

Step	Process Parameter (units)	Target	NOR	Justification/Notes
Control Method: Constant TMP and Feed Flow Control	Protein concentration, endpoint check (g/L)	100	63.4─ 110	May be directly measured or calculated from permeate or retentate volume
Recirculation	Feed pressure (psig)	25	≤ 60	NA
	Feed flow (L/min/m^2)	NA	≤ 7.0	NA
	Recirculation time (min)	5	≤ 120	NA
Recovery Flush *20 mM Histidine/Histidine-HCl, pH 5.9* Recovery Method: Buffer flush of system hold up volume	Feed pressure (psi)	15	≤ 60	NA
	Feed flow (L/min/m^2)	NA	≤ 7.0	NA
	Protein concentration, endpoint check (g/L)	85.0	63.4 ─ 110	NA
Dilution to Pre-formulated Intermediate *20 mM Histidine/Histidine-HCl, pH 5.9*	Protein concentration, PFI (g/L)	66.7	63.4 ─ 70.0	NA
	pH, (endpoint check) [a]	6.0	5.8 – 6.2	NA

[f] Process output

Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

Table 13 Concentration and Diafiltration: Density Values

Buffer	Step	Protein Concentration (mg/mL)	Density (g/mL)	Notes
50 mM sodium acetate, 200 mM sodium chloride pH 5.1	Conc. 1, final	60 – 80 (70)	1.0266	PPS Development eLN: MS01467-65
20 mM Histidine/Histidine-HCl, pH 5.9	PFI	63.4 ─ 70.0 (66.7)	1.0180	DLIMS: DEV-20240202-001 1252267
	Recovery	70 – 90 (85.0)	1.0199	PPS Development eLN: MS01467-65
	Conc. 2, final	90 – 110 (100)	1.0270	DLIMS: DEV-20240202-001 1242264

Check th version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

12 FORMULATION

Step	Process Parameter (units)	Target	NOR	Justification/Notes
EAB Spike (of PFI to FB) *20 mM histidine/histidine-HCl, 960 mM sucrose, pH* 6.0 (density 1.127g/mL)	Ratio of EAB spike to PFI (kg/kg)	0.367	0.356 – 0.378	FB/antibody intermediate: *50g/L INT-017, 20 mM histidine/histidine-HCl, 240 mM Sucrose, pH 6.0* *(Experimental* density 1.044 g/mL) *(DLIMS :DEV-20211116-001, 977335)*
	Protein concentration, endpoint check (g/L)	50.0	45 – 55	NA
	Osmolality (mOsm/kg) [a]	320	256 – 384	NA

[g] Process output

Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

13 BULK FILTRATION

Step	Process Parameter (units)	Target	NOR	Justification/Notes
General	Drug Substance container	CryoVaults		NA
Buffer Flush *EAB:* *20 mM histidine/histidine-HCl,* *960 mM sucrose,* *pH 6.0*	Buffer flush amount (L/m2)	NA	≥ 11	NA
Product Flush	Product flush amount (L/m2)	NA	≥ 1.9	NA
0.2 µm Bulk Filtration	Fill volume per container (L)	NA	4.0 – 16.0	If multiple containers are required, target equivalent volume in each.
	Antibody Intermediate storage temperature (°C)	-40	-45 – -35	mAb intermediate can be held 3 months at 2-8°C prior to shipment. mAb intermediate long term storage condition is ≤ -35° C. The antibody can be stored at a target of -40° C or -80° C depending on the available storage conditions per facility.
		-80	≤ -65	
	Integrity Test Result (PASS/FAIL) [a]	PASS	PASS	Perform per appropriate SOP. Integrity test is an in-process control that must pass for the lot to be released

Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

14 COLUMN PACKING AND TESTING

Refer to DEV000_GB_0469 for standard resins. New packing tables are listed below.

Table 14 CaptureSelect Lambda XP Column Packing and Testing

CaptureSelect Lambda XP Packing Step	Process Parameter (units)	Target	NOR/Action Limit	
			Min	Max
General Properties	Resin Name	CaptureSelect Lambda XP		
	Manufacturer	Thermo Fisher Scientific		
	Particle size (µm)	65		
	Recommended screen size (µm)	20-23		
	Composition	highly cross-linked agarose		
	Packed bed pressure limit (psi)	30		
Packing *100 mM sodium chloride*	Bed height (cm)	19	16	22
	Slurry %	50	30	70
	Compression factor	1.20	NA	NA
	Packing method	Flow/Axial	NA	NA
	Packing flow rate (cm/hr)	400	60	450
	Packing pressure (psi)	NA	NA	55
	Slurry delivery velocity (cm/hr)	400	350	450
	Adapter descent rate (mm/sec)	0.2	0.1	0.3
Qualification Test Running (equilibration/ chase/ pressure-flow) buffer: *100 mM sodium chloride* Spike buffer: *1.0 M sodium chloride*	Equilibration flow rate (cm/hr)	200	NA	220
	Equilibration volume (CV)	3.0	2.5	NA
	Spike flow rate (cm/hr)	100	90	110
	Spike volume (% of column volume)	2.0	1.5	2.5
	Chase flow rate (cm/hr)	100	90	110
	Chase volume (CV)	2.0	1.5	NA
	Asymmetry	1.00	0.80	1.84
	HETP (cm)	NA	NA	0.1
Sanitization *120 mM phosphoric acid, 167 mM acetic acid, 2.2% (v/v) benzyl alcohol* or *120 mM phosphoric acid, 167 mM acetic acid, 2.0% (v/v) benzyl alcohol*	Linear velocity (cm/hr)	200	NA	220
	Volume (CV)	3.0	2.5	6.0
	Contact time (mins)	30	29	60
Storage *2% (v/v) benzyl alcohol, 100 mM sodium acetate, pH 5.0*	Linear velocity (cm/hr)	200	NA	220
	Volume (CV)	3.0	2.5	6.0

Check th version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

15 BUFFERS AND SOLUTIONS

Buffer and solution stability studies are a commercial process definition activity. However, buffers and solutions with completed studies because of prior experience or a stability concern have stated hold times below. The studies were performed by holding buffers and solutions at ambient temperature in stainless steel containers. The studies were designed to test chemical stability for longer than the expected maximum time for routine storage. These times may be extended by an appropriate study unless otherwise noted. The studies do not address microbial control, which should be defined by the facility. Assays used to measure stability included pH and conductivity. In some cases, the hold time is measured directly (direct). In other cases, the hold time is supported by the hold time of similar buffers if the proposed buffer falls within our current buffer matrix (matrix). Please see DEV000_GB_0533 and PRO-0188988 for reference.

Table 15 Buffer and Solution Stability

Buffer Name	Recipe Table	MEDI PN/ Doc Num	Hold Time - Max (days, 15-25°C)	Unit Operation	Steps
50 mM tris, pH 7.4	PRO-0188988	8003734/ A-005378	35 (direct)	Affinity	Depth Filter Equilibration
					Equilibration 1
					Equilibration 2
					Equilibration 3
					Pre-Storage Equilibration
				AEX	Equilibration
					Chase
					Line Chase

Check th version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

Buffer Name	Recipe Table	MEDI PN/ Doc Num	Hold Time - Max (days, 15-25°C)	Unit Operation	Steps
50 mM tris, 0.5 M sodium chloride, pH 7.4	PRO-0188988	8003737/ A-005380	35 (matrix)	Affinity	Wash
				AEX	Pre-Equilibration
					Salt Strip
					Pre-Storage
				CEX	Salt Strip
75 mM sodium acetate, pH 3.8	Table 16	TBD	35 (matrix)	Affinity	Elution
					Line chase
WFI	NA	NA	NA	Affinity	WFI Flush
					Pre-Storage Equilibration
				AEX	Pre-Storage Equilibration
100 mM acetic acid	PRO-0188988	A-005041	30 (direct)	Affinity	Strip 1
25 mM sodium hydroxide	PRO-0188988	8007153	NA	Affinity	Strip 2
2.0% (v/v) benzyl alcohol, 167 mM acetic acid	Table 17	Table 17	30 (matrix)	Affinity	Strip 3
2.0% (v/v) benzyl alcohol, 120 mM phosphoric acid, 167 mM acetic acid	PRO-0188988	8008420	7 (direct)	Affinity	Strip 4

Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

Buffer Name	Recipe Table	MEDI PN/ Doc Num	Hold Time - Max (days, 15-25°C)	Unit Operation	Steps
0.1 N sodium hydroxide	PRO-0188988	A0014-D/ A0014-D	180 (matrix)	CEX	Storage
2% (v/v) benzyl alcohol, 100 mM sodium acetate, pH 5.0	PRO-0188988	8002256/ A-005212	30 (direct)	Affinity	Storage
				AEX	Storage
1 M acetic acid	PRO-0188988	A-005168/ A-005168	77 (direct)	Viral Inactivation	Acid Titration
				AEX Product Adjustment	Adjustment
1 M tris base	PRO-0188988	A0101-D/ A0101-D	35 (matrix)	Viral Inactivation	Base Titration
50 mM sodium acetate, pH 5.0	PRO-0188988	A-005097/ A-005097	35 (direct)	CEX	Equilibration 1
					Equilibration 2
35 mM sodium phosphate, pH 6.9	Table 18	Table 18	30 (matrix)	CEX	Wash
50 mM sodium acetate, 200 mM sodium chloride, pH 5.1	Table 19	Table 19	30 (matrix)	CEX	Elution
					Line chase
				Virus Filtration	Equilibration
					Chase
1.0 N sodium hydroxide	PRO-0188988	A0191-D/ A0191-D	180 (direct)	AEX	NaOH Strip
				CEX	NaOH Strip
20 mM Histidine/Histidine-HCl, pH 5.9	PRO-0188988	8006723	NA [a]	Concentration and Diafiltration	Equilibration
					Diafiltration
					Recovery Flush
					Dilution to PFI
20 mM Histidine/Histidine-HCl, 960 mM sucrose, pH 6.0	PRO-0188988	8005513/ A005465	NA [a]	Antibody intermediate Formulation	Formulation of PFI to FB

Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

Table 16 75 mM sodium acetate, pH 3.8

MEDI PN/ Document No.	TBD		
BPD Console No.	B-851		
Density Corrected	Yes		
Specification	**Design Target**	**NOR/Action Limit**	
		Min	**Max**
pH	3.8	3.6	4.0
Conductivity (mS/cm at 25°C) (FIO)	NA	Run and record	
Density (g/mL)	0.9985	NA	NA
Chemical	**MEDI PN**	**Amount by Volume**	**Amount by Weight**
Sodium acetate, trihydrate	4101342 6000326	NA	1.165 g/kg
Acetic acid, glacial	4101341 6000325	NA	3.984 g/kg
WFI	NA	NA	0.995 kg/kg
Total		1000 g/kg	

Table 17 2.0% (v/v) benzyl alcohol, 167 mM acetic acid

MEDI PN/ Document No.	TBD		
BPD Console No.	B-842		
Density Corrected	Yes		
Specification	**Design Target**	**NOR/Action Limit**	
		Min	**Max**
pH	2.7	2.5	2.9
Conductivity (mS/cm at 25°C) (FIO)	NA	Run and record	
Density (g/mL)	0.9985	NA	NA
Chemical	**MEDI PN**	**Amount by Volume**	**Amount by Weight**
Benzyl alcohol	4102670	NA	20.8 g/kg
Acetic acid, glacial	4101341 6000325	9.51 mL/kg	10.0 g/kg
WFI	NA	NA	0.969 kg/kg
Total		1000 g/kg	

Check th version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

Table 18 35 mM sodium phosphate, pH 6.9

MEDI PN/ Document No.	TBD		
BPD Console No.	B-850		
Density Corrected	Yes		
Specification	**Design Target**	**NOR/Action Limit**	
		Min	**Max**
pH	6.9	6.8	7.0
Conductivity (mS/cm at 25°C) (FIO)	NA	Run and record	
Density (g/mL)	1.0016	NA	NA
Chemical	**MEDI PN**	**Amount by Volume**	**Amount by Weight**
Sodium Phosphate, Dibasic, Heptahydrate	4101092	NA	4.797 g/kg
Sodium Phosphate, Monobasic, Monohydrate	4101091	NA	2.297 g/kg
WFI	NA	NA	0.993 kg/kg
Total		1000 g/kg	

Table 19 50 mM sodium acetate, 200 mM sodium chloride, pH 5.1

AZ PN/ Document No.	TBD		
BPD Console No.	B-387		
Density Corrected	Yes		
Specification	**Design Target**	**NOR/Action Limit**	
		Min	**Max**
pH	5.1	4.9	5.2
Conductivity (mS/cm at 25°C)	22.0	Run and record (FIO)	
Density (g/mL)	1.01	NA	NA
Chemical	**MEDI PN**	**Amount by Volume**	**Amount by Weight**
Sodium acetate, trihydrate	4101342	NA	4.87 g/kg
Acetic acid, glacial	4101341	0.78 mL/kg	NA
Sodium chloride	6000324	NA	11.57 g/kg
WFI	NA	NA	0.983 kg/kg
Total		1000 g/kg	

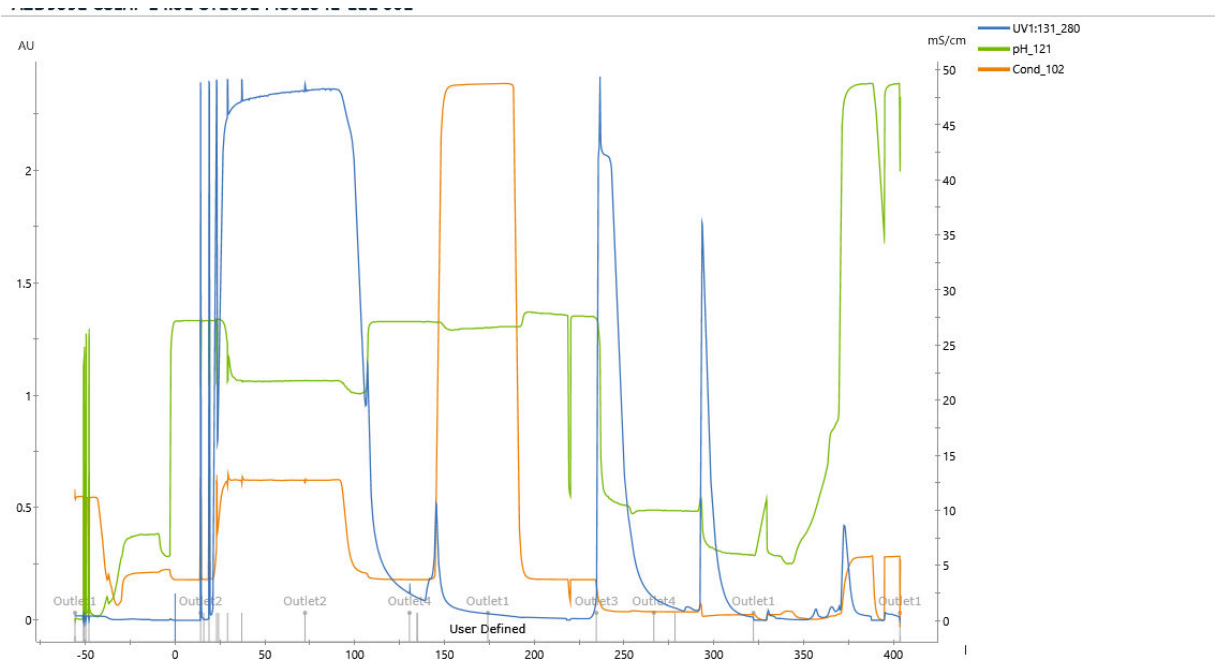
Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

16 REFERENCE CHROMATOGRAMS

Sample chromatograms representing the three chromatography steps used in the purification process are presented below. It is preferred to provide the first cycle coming out of storage and the largest run scale.

Table 20	**CaptureSelect Lambda XP Chromatogram**
Resin:	CaptureSelect Lambda XP
Column Dimensions:	14.5L (30cm diameter)
Notes:	200L batch (SP24-010)

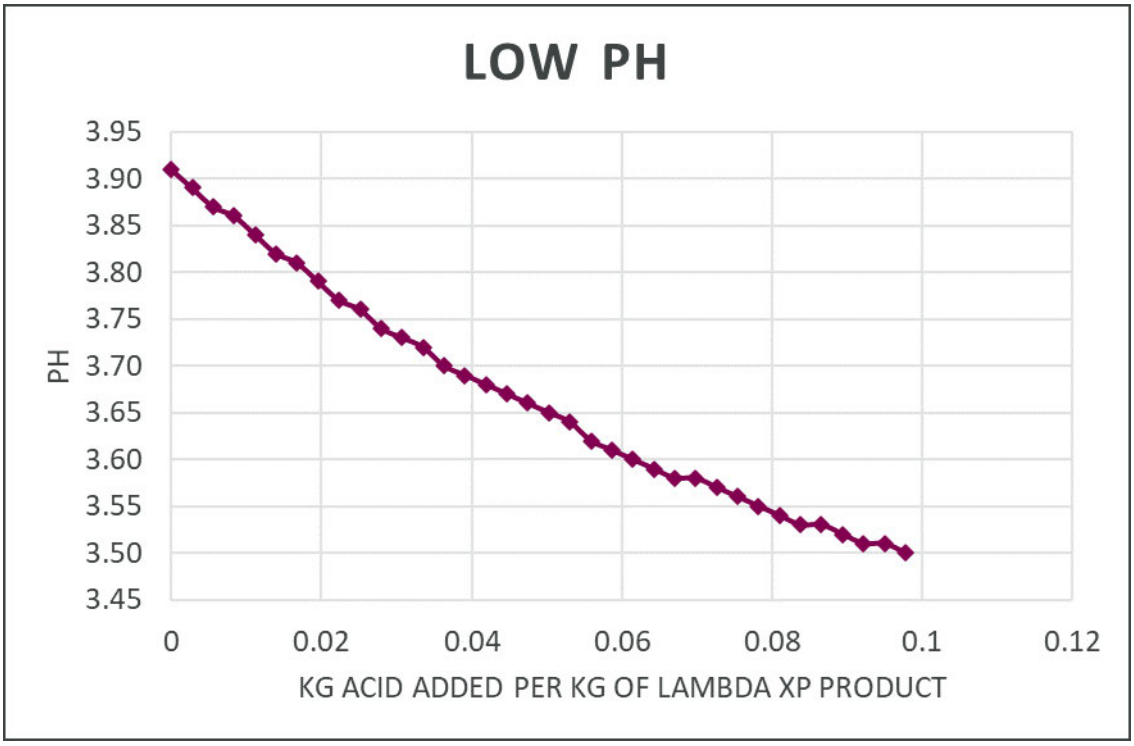
Figure 1 CaptureSelect Lambda XP Chromatogram



Check th version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

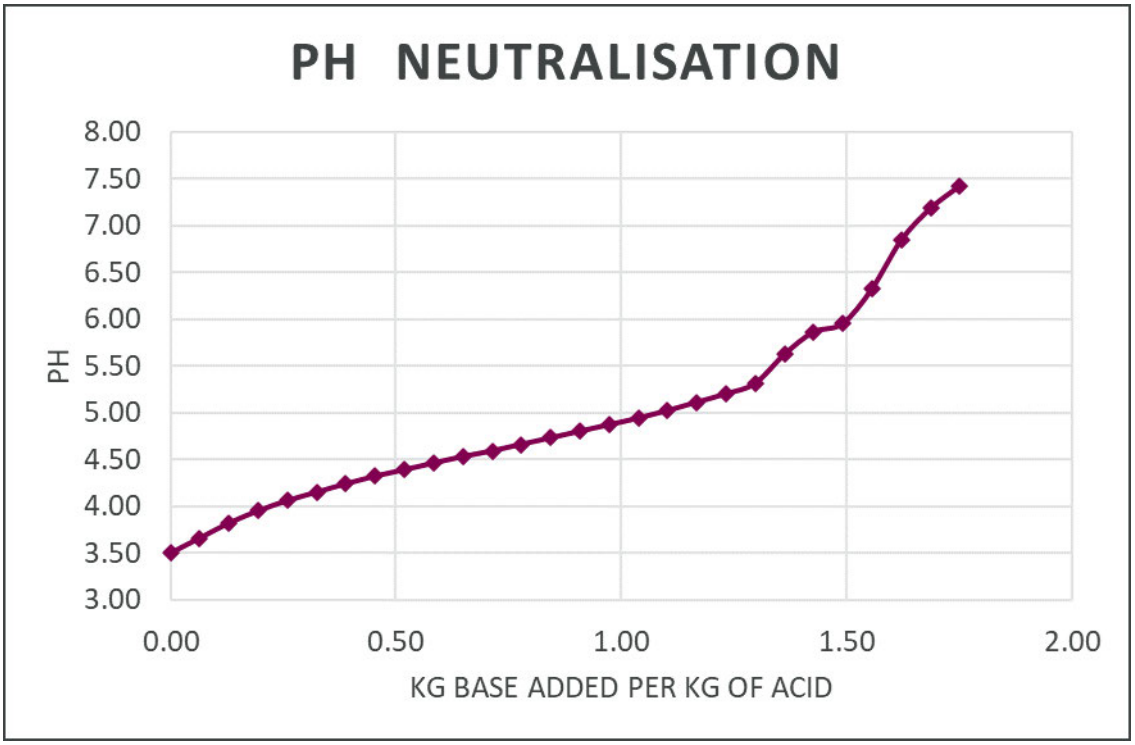
Table 21	**Virus Inactivation Titration Curves**
Notes:	200L batch SP24-010

Figure 2 Acid Titration



Check th version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

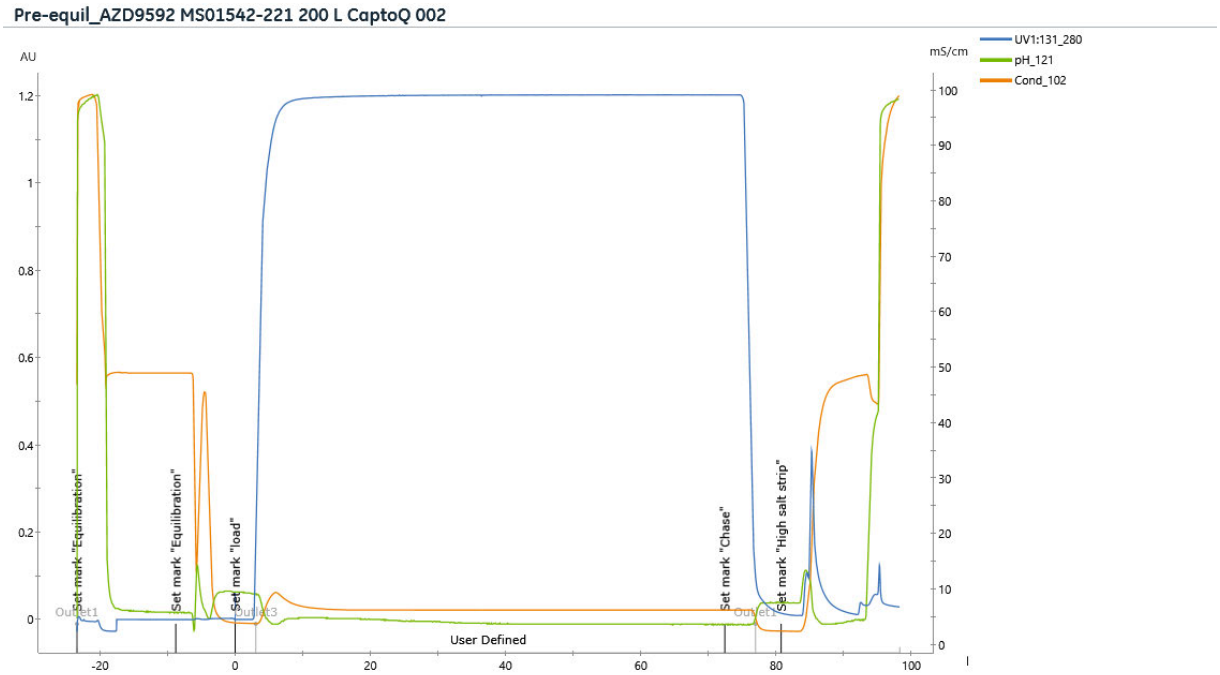
Figure 3 Base Titration



Check th version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

Table 22	**AEX Chromatogram**
Membrane:	Capto Q
Column Dimensions:	2.925L (14cm diameter)
Notes:	200L batch SP24-010

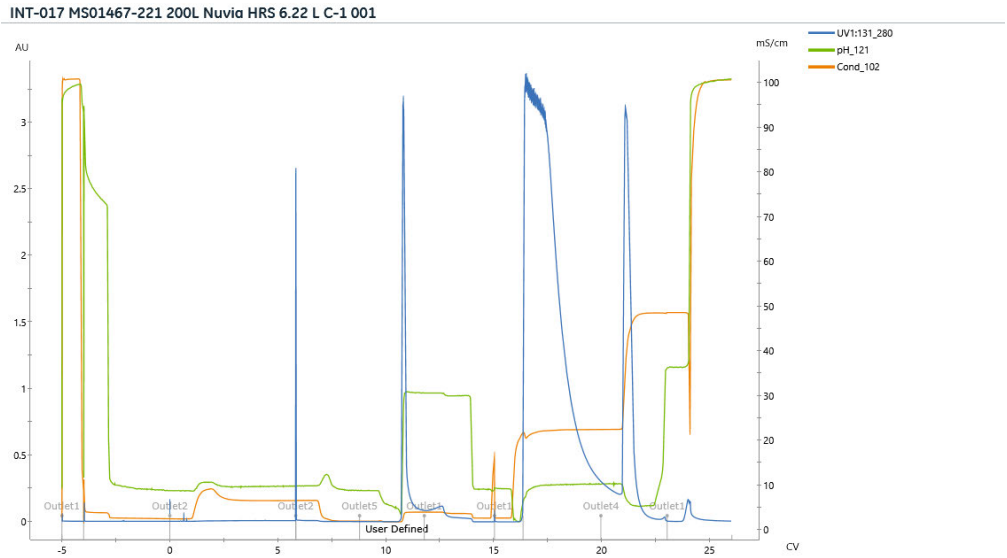
Figure 4 AEX Chromatogram



Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

Table 23	**CEX Chromatogram**
Resin:	Nuvia HRS
Column Dimensions:	6.22L (20cm diameter)
Notes:	200L batch SP24-010

Figure 5 CEX Chromatogram



Check th version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

17 PROCESS CHANGES

Table 24 lists changes in the INT-017 Purification Process Description from Edition 000 to Edition 002. These changes warrant a process number change from Process 0 to Process 1.

Table 24 Process 0 to Process 1 Change Summaries and Justifications

Unit Operation	Step	Parameter	Original	New	Summary/ Justification
Affinity Chromatography	General	Media	**ED 000** [a]: Protein A	**ED 001**: Lambda Fab Select	Affinity resin for duet Mab.
	Elution	Buffer	**ED000:** 25 mM sodium acetate, pH 3.6	**ED001:** 75 mM sodium acetate, pH 3.5	Optimized for product quality.
	Sanitization	Buffer	**ED 000**: 0.1 N sodium hydroxide	**ED 001**: 120 mM phosphoric acid, 167 mm acetic acid, 2.2 % (v/v) benzyl alcohol	Optimized for resin
	Storage	Buffer	**ED 000**: 2% (v/v) benzyl alcohol, 100 mM sodium acetate, pH 5.0	**ED 001**: 20% Ethanol	Optimized for resin
Viral Inactivation	Incubation time	NOR	**ED 000**[a]: 30-120 min	**ED 001**: 60 min-48 hr	Viral clearance
CEX	General	Media	**ED000:** Poros HS50	**ED 001:** Nuvia HRS	Optimization for product quality
LambdaFab Select	General	POD Filter	**ED001:** No POD filter	**ED 002:** Millistak+ HC POD filters - X0HC grade	Added to increase guard filter capacity and decrease pressure issues
Virus Filtration	Total volumetric load	NOR	**ED001:** TBD	**ED002: <=1000L/m^2**	Updated based on VC study performance

[a] Edition 000 and Process 0 represent the platform process template; therefore, changes from Edition 000 do not represent true process changes but are listed here simply for tracking purposes and ease of review. Edition 001 and Process 1 is the first recognized Process Description.

Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

Table 25 lists changes in the INT-017 Purification Process Description from Edition 002 to Edition 003. These changes warrant a process number change from Process 1 to Process 2.

Table 25 Process 1 to Process 2 Change Summaries and Justifications

Unit Operation	Step	Parameter	Original	New	Summary/ Justification
Affinity Chromatography	General	Media	**ED002:** LambdaFabSelect	**ED003**: CaptureSelect Lambda XP	New resin selected to increase binding capacity.
	Equilibration 1	Linear velocity (cm/hr)	**ED002:** Target: 300 AL: 285 — 315 NOR: ⩽ 330	**ED003:** Target: 200 NOR: ⩽ 220	Optimized for CaptureSelect Lambda XP
	Load	Protein load (g/L resin)	**ED002:** Target: NA AL: 10 — 20 NOR: ⩽ 20	**ED003:** Target: NA NOR: 15 — 30	Optimized for CaptureSelect Lambda XP
	Equilibration 2	Linear velocity (cm/hr)	**ED002:** Target: 300 AL: 285 — 315 NOR: ⩽ 330	**ED003:** Target: 200 NOR: ⩽ 220	Optimized for CaptureSelect Lambda XP
	Wash	Linear velocity (cm/hr)	**ED002:** Target: 300 AL: 285 — 315 NOR: ⩽ 330	**ED003:** Target: 200 NOR: ⩽ 220	Optimized for CaptureSelect Lambda XP
	Equilibration 3	Linear velocity (cm/hr)	**ED002:** Target: 300 AL: 285 — 315 NOR: ⩽ 330	**ED003:** Target: 200 NOR: ⩽ 220	Optimized for CaptureSelect Lambda XP
	Elution	Buffer	**ED002:** 75 mM sodium acetate, pH 3.5	**ED003:** 75 mM sodium acetate, pH 3.8	Optimized for CaptureSelect Lambda XP

Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

Table 25 lists changes in the INT-017 Purification Process Description from Edition 002 to Edition 003. These changes warrant a process number change from Process 1 to Process 2.

Table 25 Process 1 to Process 2 Change Summaries and Justifications

Unit Operation	Step	Parameter	Original	New	Summary/ Justification
Affinity Chromatography	Elution	Linear velocity (cm/hr)	**ED002:** Target: 150 AL: 142 — 158 NOR: ≤ 330	**ED003:** Target: 150 NOR: ≤ 165	Optimized for CaptureSelect Lambda XP
	Strip 1	Linear velocity (cm/hr)	**ED002:** Target: 300 AL: 285 — 315 NOR: ≤ 330	**ED003:** Target: 200 NOR: ≤ 220	Optimized for CaptureSelect Lambda XP
	NaOH Strip	Linear velocity (cm/hr)	**ED002:** Target: 300 AL: 285 — 315 NOR: ≤ 330	**ED003:** Target: 200 NOR: ≤ 220	Optimized for CaptureSelect Lambda XP
	Strip 2	Buffer	**ED002:** 120 mM phosphoric acid, 167 mM acetic acid, 2.2 % (v/v) benzyl alcohol	**ED003:** 2.0% (v/v) benzyl alcohol, 167 mM acetic acid	Optimized for CaptureSelect Lambda XP
	Sanitization Hold	Contact time (minutes)	**ED002:** Target: 35 AL: 30 — 60 NOR: 30 — 120	**ED003:** Target: 30 NOR: 29 — 60	Optimized for CaptureSelect Lambda XP
	Pre-Storage Flush	Buffer	**ED002:** 50 mM tris, 0.5 M sodium chloride, pH 7.4	**ED003:** 50 mM tris, pH 7.4 OR WFI	Optimized for CaptureSelect Lambda XP

Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

Table 25 lists changes in the INT-017 Purification Process Description from Edition 002 to Edition 003. These changes warrant a process number change from Process 1 to Process 2.

Table 25 Process 1 to Process 2 Change Summaries and Justifications

Unit Operation	Step	Parameter	Original	New	Summary/ Justification
Affinity Chromatography	Pre-Storage Flush	Linear velocity (cm/hr)	**ED002:** Target: 300 AL: 285 — 315 NOR: ⩽ 330	**ED003:** Target: 200 NOR: ⩽ 220	Optimized for CaptureSelect Lambda XP
		Volume (CV)	**ED002:** Target: 3.0 AL: 2.9 — 4.0 NOR: ⩾ 2.5	**ED003:** Target: 1.0 NOR: ⩾ 0.5	Optimized for CaptureSelect Lambda XP
	Storage	Linear velocity (cm/hr)	**ED002:** Target: 300 AL: 285 — 315 NOR: ⩽ 330	**ED003:** Target: 200 NOR: ⩽ 220	Optimized for CaptureSelect Lambda XP
Virus Inactivation	NA	Duration (minutes)	**ED002:** Target: 65 AL: 60 — 240 NOR: 60 min — 48 hr	**ED003:** Target: 65 NOR: 60 — 240	Updated to align with VC study design
CEX Chromatography	Wash	Buffer	**ED002:** 20 mM sodium phosphate, 22 mM sodium chloride, pH 6.9	**ED003:** 35 mM sodium phosphate, pH 6.9	Process 2 optimization
	Elution	Buffer	**ED002:** 50 mM sodium acetate, 200 mM sodium chloride pH 5.0	**ED003:** 50 mM sodium acetate, 200 mM sodium chloride pH 5.1	Process 2 optimization

Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

Table 25 lists changes in the INT-017 Purification Process Description from Edition 002 to Edition 003. These changes warrant a process number change from Process 1 to Process 2.

Table 25 Process 1 to Process 2 Change Summaries and Justifications

Unit Operation	Step	Parameter	Original	New	Summary/ Justification
Virus Filtration	Equilibration	Buffer	**ED002:** 50 mM sodium acetate, 200 mM sodium chloride pH 5.0	**ED003:** 50 mM sodium acetate, 200 mM sodium chloride pH 5.1	Process 2 optimization
	Chase	Buffer	**ED002:** 50 mM sodium acetate, 200 mM sodium chloride pH 5.0	**ED003:** 50 mM sodium acetate, 200 mM sodium chloride pH 5.1	Process 2 optimization
Concentration and Diafiltration	Concentration 1	Membrane protein load (g/m2)	**ED002:** Target: NA AL: 500 — 1000 NOR: ≤ 1100	**ED003:** Target: NA NOR: ≤ 1000	Updated to align with new PD template without Alert Limits
Formulation	EAB Spike (of PFI to UB)	Protein concentration, endpoint check (g/L)	**ED002:** Target: 50.0 AL: 47.5 — 52.5 NOR: NA	**ED003:** Target: 50.0 NOR: 45 — 55	Updated to align with new PD template without Alert Limits
		Osmolality (mOsm/kg)	**ED002:** Target: 320 AL: 256 — 384 NOR: NA	**ED003:** Target: 320 NOR: 256 — 384	Updated to align with new PD template without Alert Limits
Bulk Filtration	0.2 µm Bulk Filtration	mAb Intermediate container	**ED002:** HDPE bottles	**ED003:** Cryovaults	Process 2 optimization

Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

Table 25 lists changes in the INT-017 Purification Process Description from Edition 002 to Edition 003. These changes warrant a process number change from Process 1 to Process 2.

Table 25 Process 1 to Process 2 Change Summaries and Justifications

Unit Operation	Step	Parameter	Original	New	Summary/ Justification
Bulk Filtration	0.2 µm Bulk Filtration	Fill volume per container	**ED002:** Target: 70% AL: NA NOR: ≤ 90	**ED003:** Target: NA NOR: 4.0 — 16.0	Updated to align with new mAb Intermediate container

[b] Edition 000 and Process 0 represent the platform process template; therefore, changes from Edition 000 do not represent true process changes but are listed here simply for tracking purposes and ease of review. Edition 1.00 and Process 1 is the first recognized Process Description.

Check th[redacted] version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

18 EDITION HISTORY

Edition	Reason
1.0	• New document
2.0	• Updated LambdaFab Select Chromatography product filtration section to have an X0HC grade POD filter and increased volumetric load for Load Filter 2 to 3000L/m2. Added note to add depth filter holdup volume to the target. Updated process changes section to include POD filter addition. Updated Total volumetric load NOR from TBD to <=1000 to reflect results of VC study. Updated process changes section to reflect Total Volumetric Load NOR update.
3.0	• Changed Affinity storage buffer.
4.0	• Updated Process Description template to align with new template. • Updated document throughout to align with Process 2. • See Table 25 for summary of changes from Process 1 to Process 2.

Check th version of the document before use.
Printed by [redacted] on 01 Oct 2024 08:17AM GMT-04:00

Document Approvals

Business Approval	[redacted]
Business Approval	[redacted]
Business Approval	[redacted]
Business Approval	[redacted]
Business Approval	[redacted]